Structural Aliasing Analysis in Time-Series Forecasting Using Chronos

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Executive Summary

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Time-Series Forecasting

Time-series forecasting underpins operational decision-making wherever a system's future state must be anticipated from its past, from grid load to industrial condition monitoring: projecting historical sequences into future horizons is what allows a control policy to be proactive rather than reactive[1].

During the preparation of this work, the authors used AI-based tools, including GPT-5.6 Sol, Gemini 3.7 Flash, Sonnet 5, and Opus 5, to support language refinement and improve clarity. All generated or suggested content was carefully reviewed, revised where necessary, and validated by the authors, who assume full responsibility for the final content of this work.

Evolution of Forecasting Methodologies

Classical time-series analysis relies on statistical models such as AutoRegressive Integrated Moving Average (ARIMA) and Exponential Smoothing (ETS)[1]. These are interpretable and robust on stable low-dimensional data, but they assume linearity and stationarity and are fitted to one series at a time.

Deep learning relaxed those assumptions: recurrent networks and their gated variants[2] model non-linear dynamics through a hidden state, and training jointly across many related series established the global forecasting paradigm[3]. Transformers then replaced the recurrent bottleneck with self-attention[4], computing dependencies between distant time steps directly. Two costs follow for sampled signals: attention scales quadratically with sequence length, and a single time step is a weak unit of meaning. Patching addresses both, segmenting the series into contiguous sub-sequences and embedding each as one token, which shortens the sequence and gives every token a locally semantic support[5].

Time-series foundation models sit at the end of this line. Chronos treats time-series values as discrete tokens obtained by scaling and quantisation[6]; its successor Chronos-Bolt replaces quantisation with the patching scheme above and a direct multi-quantile head[7].[1] Patching is therefore not an incidental implementation detail, but the design choice that makes this class of models practical at scale.

Classic Aliasing

Whatever the architecture downstream, predictive fidelity is bounded by the digital representation of the signal. The governing limit is the Nyquist-Shannon sampling theorem[9]: a band-limited continuous-time signal is perfectly reconstructible only if sampled uniformly at a rate f_s strictly greater than twice its highest frequency component $f_{\max}$,

$$f_s > 2f_{\max}. \tag{1}$$

The threshold $f_N = f_s/2$ is the Nyquist frequency. When the criterion is violated, through an insufficient sampling rate or a missing anti-aliasing filter, components above f_N fold back into the lower spectrum and become indistinguishable from genuine low-frequency content[10]. This is *aliasing*, and it is irreversible: the information is destroyed at acquisition, not at inference. Respecting Equation 1 is consequently taken as sufficient for a discrete sequence to represent its source faithfully. This report asks whether that sufficiency survives patch-based tokenization.

Application Domain and Scope

The reference setting is a designed scenario: a photovoltaic array supplying an internal load, a local storage battery and the national grid, with a control agent performing load balancing, battery dispatch, state monitoring and anomaly detection on short-horizon forecasts. Measured photovoltaic and weather telemetry are taken from the SolarTech Lab dataset[11]; the concepts, states and relations of the scenario are formalised as an OWL2 ontology in Phase locking and Aliasing.

The target architecture is Chronos-Bolt-Tiny, with default patch size $P = 16$, stride $S = 16$, context window $T = 2048$, four encoder and four decoder blocks, and a direct quantile projection head[12]. The objective is to test the assumption that such a model preserves the frequency content of Nyquist-compliant signals, and to characterise how f_s, P and S interact in determining when it does not[13].

[1] The vendor reports up to a 250× inference speed-up over the original Chronos[8]; no peer-reviewed description of Chronos-Bolt has been published.

Patch-Based Tokenization: a Formal Description

Let $\mathbf{x} = \{x[n]\}_{n\in\mathbb{Z}}$ be a discrete-time signal representing the input time series. We define the patch-based tokenization operator $\mathcal{T}_{P,S}$ with patch length $P \in \mathbb{N}^+$ and stride $S \in \mathbb{N}^+$ as a mapping that transforms the signal $\mathbf{x}$ into a sequence of vector-valued tokens $\mathbf{T} = \{\mathbf{t}_k\}_{k\in\mathbb{Z}}$, where each token $\mathbf{t}_k \in \mathbb{R}^P$.

The k-th token is defined as

$$\mathbf{t}_k = \mathcal{T}_{P,S}(\mathbf{x})[k] = \begin{bmatrix} x[kS] \\ x[kS+1] \\ \vdots \\ x[kS+P-1] \end{bmatrix}. \tag{2}$$

Using element-wise indexing notation,

$$\mathbf{t}_k[m] = x[kS+m], \qquad m \in \{0,\dots,P-1\}. \tag{3}$$

Before examining the degeneracies of this operator it is worth fixing what it does not do. Whenever $S \le P$, every sample of the input appears in at least one token and can be read back from it: taking $k = \lfloor n/S \rfloor$ and $m = n \bmod S \le S-1 \le P-1$ in Equation 3 gives

$$x[n] = \mathbf{t}_{\lfloor n/S \rfloor}[n \bmod S], \qquad S \le P. \tag{4}$$

Complete raw patching with $S \le P$ is therefore injective: the token sequence determines the signal, and no frequency below Nyquist is destroyed by tokenization itself. Everything that follows concerns redundancy in the token sequence rather than loss of information, the single exception being the $S > P$ regime of Case 3 below, where the operator does discard samples.

Phase locking and Aliasing

Consider a continuous-time periodic signal sampled uniformly at sampling frequency f_s, yielding the discrete-time sequence $x[n]$. We isolate a fundamental harmonic component and model it as

$$x[n] = \sin\left(2\pi \frac{f}{f_s} n + \phi\right), \tag{5}$$

where f denotes the signal frequency satisfying the Nyquist criterion $f < \frac{f_s}{2}$, and $\phi \in [0, 2\pi)$ is an arbitrary phase offset.

Phase-locking occurs when the structural phase profile of the signal repeats exactly under a translational shift equal to the stride S. Evaluating a stride translation yields

$$x[n+S] = \sin\left(2\pi \frac{f}{f_s}(n+S) + \phi\right) = \sin\left(2\pi \frac{f}{f_s} n + 2\pi \frac{fS}{f_s} + \phi\right). \tag{6}$$

For exact alignment between the signal and the tokenization operator, the phase accumulation over one stride must correspond to an integer number of 2π-periods

$$2\pi \frac{fS}{f_s} = 2\pi c, \qquad c \in \mathbb{N}^+. \tag{7}$$

As a result, the signal exhibits exact stride-period translational invariance

$$x[n+S] = \sin\left(2\pi \frac{f}{f_s} n + 2\pi c + \phi\right) = \sin\left(2\pi \frac{f}{f_s} n + \phi\right) = x[n]. \tag{8}$$

The same derivation can be applied taking into account translational shifts equal to the patch size P, yielding a similar phase-locking condition

$$2\pi \frac{fP}{f_s} = 2\pi k \Rightarrow x[n+P] = x[n], \qquad k \in \mathbb{N}^+. \tag{9}$$

Solving Equation 7 and Equation 9 for f yields the phase-locking frequency class $\mathcal{F}_{\text{lock}}$, which is defined as the set of frequencies producing self-repetition over one stride or over one patch.

$$\mathcal{F}_{\text{lock}} = \left\{ f \in \mathbb{R}^+ \;\middle|\; f = c\frac{f_s}{S} \vee f = k\frac{f_s}{P},\ c, k \in \mathbb{N}^+,\ f < \frac{f_s}{2} \right\}. \tag{10}$$

It is convenient to express both conditions in units that do not depend on the sampling rate. Introducing the cycles-per-stride and cycles-per-patch ratios

$$\text{cps} = \frac{fS}{f_s}, \qquad \text{cpp} = \frac{fP}{f_s}, \tag{11}$$

Equation 7 and Equation 9 read simply $\text{cps} \in \mathbb{N}^+$ and $\text{cpp} \in \mathbb{N}^+$: $\mathcal{F}_{\text{lock}}$ is the set of frequencies completing a whole number of cycles within one stride or within one patch. This normalisation makes the two branches directly comparable across geometries and across sampling rates.

Neither branch is restricted to sinusoidal signals. Both conditions generalise to an arbitrary discrete-time signal invariant under a translation equal to the stride or to the patch,

$$x[n+S] = x[n] \qquad \text{or} \qquad x[n+P] = x[n], \qquad \forall n, \tag{12}$$

but only the first bears on consecutive patches. Tokens begin at multiples of the stride, so the step from token k to token $k+j$ is a shift of jS samples. Substituting Equation 3 and reducing by the stride condition at $j=1$ gives

$$\mathbf{t}_{k+1}[m] = x[(k+1)S + m] = x[kS + m + S] = x[kS + m] = \mathbf{t}_k[m], \tag{13}$$

and since this holds for every component,

$$\mathbf{t}_{k+1} = \mathbf{t}_k. \tag{14}$$

The patch condition reduces the same shift only when P divides jS. At $j=1$ that requires $P \mid S$, which under $S \leq P$ occurs only at $S = P$, so consecutive tokens are left distinct. It does hold at the smallest j for which $P \mid jS$, namely $j = P/\gcd(P,S)$, giving

$$\mathbf{t}_{k+j}[m] = x[(k+j)S + m] = x[kS + m + jS] = x[kS + m] = \mathbf{t}_k[m], \qquad j = \frac{P}{\gcd(P,S)}, \tag{15}$$

and therefore

$$\mathbf{t}_{k+j} = \mathbf{t}_k. \tag{16}$$

Patch-periodicity thus makes tokens equal as well, but at a spacing of j rather than between neighbours. The two conditions coincide on consecutive patches only in the non-overlapping geometry, where $j = 1$.

The derivations above show that some translations leave a signal unchanged; by Equation 4 this repetition is not, by itself, evidence that two distinct signals become indistinguishable to the model. Structural aliasing is therefore treated as an emergent, falsifiable property of the trained representations[13]: it is diagnosed by readability rather than by a distance between raw tokens, and it is present when the signal's frequency can no longer be recovered from the model's internal state at a locked frequency as well as it can at neighbouring, non-locked ones. $\mathcal{F}_{\text{lock}}$ is accordingly a set of candidate frequencies, derived from the geometry alone; whether the candidates are realised is an empirical question.

Analysis of Structural Redundancies and Invariances

Equation 10 is the union of two distinct families,

$$\mathcal{F}_S = \left\{ c\frac{f_s}{S} \right\}_{c\in\mathbb{N}^+}, \qquad \mathcal{F}_P = \left\{ k\frac{f_s}{P} \right\}_{k\in\mathbb{N}^+}, \qquad \mathcal{F}_{\text{lock}} = (\mathcal{F}_S \cup \mathcal{F}_P) \cap \left[0, \tfrac{f_s}{2}\right), \tag{17}$$

which coincide if and only if $S = P$, and which act on the token sequence in different ways. A frequency in $\mathcal{F}_S$ satisfies the stride condition, so by Equation 13 every token equals its predecessor: the sequence becomes one window repeated, and the number of distinct tokens drops to one however long the context. A frequency in $\mathcal{F}_P$ has a period $T_0 = f_s/f$ dividing P, so it satisfies the patch condition and by Equation 16 the sequence instead repeats every $j = P/\gcd(P, S)$ tokens, cycling through at most that many distinct vectors. Each token still holds a whole number of cycles, but neighbours read that waveform from different starting points and coincide only when $j = 1$. The two branches therefore differ in what they repeat rather than in whether they repeat: the stride branch acts between neighbouring tokens, the patch branch on the sequence as a whole.

Neither removes information. For $S \leq P$ both are regimes of Equation 4, so the signal remains recoverable and two inputs with the same token sequence are the same signal on the observed support. Both repetitions belong to the raw tokens, not to the learned representation; treating either as an invariance of the model would require showing that the patch embedding is itself shift-equivariant, which is neither established here nor assumed.

Case 1: Equal Patch Size and Stride ($P = S$) When patches are contiguous and non-overlapping the two branches coincide, $\mathcal{F}_S = \mathcal{F}_P$, and $j = P/\gcd(P, P) = 1$, so Equation 16 collapses onto Equation 14. This is the only geometry in which both conditions act on consecutive patches, and every locked frequency then gives

$$\mathbf{t}_k = \mathbf{t}_0, \qquad \forall k. \tag{18}$$

This is the most redundant geometry of the family: the only one in which every locked frequency reduces the token sequence to a single repeated vector, leaving no residual structure across tokens. It is also the default configuration of the target architecture, which operates at $P = S = 16$.

Case 2: Overlapping Windows ($S < P$) When the stride is smaller than the patch length the two branches separate. Since $f_s/S > f_s/P$, the stride comb is the sparser of the two: strictly below Nyquist it contributes $\lceil S/2 \rceil - 1$ members against the $\lceil P/2 \rceil - 1$ of the patch comb, and whenever S divides P the stride comb is a subset of the patch comb. Most locked frequencies in this regime therefore belong to $\mathcal{F}_P \setminus \mathcal{F}_S$, where consecutive tokens remain distinct and the sequence merely repeats with period $j > 1$.

Overlap does not weaken the exact condition. The derivation of Equation 13 makes no use of the relation between P and S, so at $f \in \mathcal{F}_S$ consecutive tokens are still exactly equal. What overlap changes is the behaviour around a lock: adjacent windows share $P - S$ samples by construction, so two consecutive tokens are constrained to agree on their common support regardless of frequency, and the transition into repetition is correspondingly smoother. Reducing the stride at fixed patch size therefore raises the lowest member of $\mathcal{F}_S$, thins the stride comb below Nyquist, and softens the local shape of the effect without removing it.

Case 3: Disjoint Windows ($S > P$) When the stride exceeds the patch length the windows leave $S - P$ unobserved samples between consecutive patches, and Equation 4 fails: $\mathcal{T}_{P,S}$ is non-injective for every f, since signals differing only on the gaps share a token sequence. The loss is one of discarded samples rather than of tokenization geometry, so these configurations are excluded by construction from the experimental design.

What the geometry does and does not establish The three cases can be summarised in four statements, of which only the last is an empirical claim. $\mathcal{F}_S$ collects the stride-lock candidates, at which consecutive raw tokens repeat. $\mathcal{F}_P$ collects the frequencies whose period divides the patch, at which the sequence repeats every $P/\gcd(P, S)$ tokens while neighbours stay distinct unless $S = P$. Their union $\mathcal{F}_{\text{lock}}$ is therefore a statement about candidate geometry, not a proof of aliasing: for $S \leq P$ the raw token sequence remains a lossless encoding of the input at every one of these frequencies. Structural aliasing is the separate, representation-level proposition that a model trained on such sequences nevertheless fails to make those frequencies readable from its internal state. It is a property of the learned representation rather than of the operator, and one that can only be established empirically. The hypotheses that test it are formulated in Bayesian Analysis.

Ontology Domain Description

The ontology reproduced in Appendix A closes the loop from the signal to the decision. It represents the photovoltaic micro-grid, its telemetry and a single control agent, and it makes the tokenisation geometry a first-class object rather than a property of the model: a `pv:TimeSeriesSignal` has a `pv:PatchedRepresentation` carrying `pv:patchSize`, `pv:stride` and a derived `pv:overlapRatio`, and one signal may hold several, one per (P, S) under evaluation. Since the lock conditions are arithmetic on those same fields, the set of candidates is derivable from the knowledge base rather than hard-coded beside it, which is what lets an agent recompute it when a configuration or a sampling interval changes.

The point of contact with the Bayesian analysis is the following. An `pv:ObservabilityAssessment` does not derive a state from the geometry alone. It records the estimand, its effect size, its credible interval and its posterior probability, and assigns StructurallyAliased only to a frequency that is a lock candidate and also carries confirmed evidence of loss, $X = Z \wedge Y$ in the notation of Appendix A. Geometry alone yields a candidate, never a verdict. Because the confirmation threshold exceeds one half, confirmed loss and confirmed preservation cannot both hold. A frequency for which preservation is positively confirmed $(\neg X \wedge R)$ is assigned Observable; everything else is assigned PartiallyObservable: an unfinished test, a credible interval crossing the threshold, or conflicting metrics. That is the inconclusive verdict of Table 3 carried into the semantics, and its consequence is operational rather than documentary, because the assessment selects the control mode. Confirmed aliasing forces the safe mode and suspends forecast-driven battery decisions; a merely partially observable assessment reduces the battery cap and lowers decision confidence. Uncertainty about the representation is therefore propagated to the actuator instead of being resolved by assumption.

Bayesian Analysis

Bayesian inference is used to test the structural-aliasing hypotheses while making the analysis assumptions and uncertainty explicit: prior assumptions are stated for each estimand, posterior uncertainty is quantified for every estimated effect, and competing hypotheses and the mitigation claim are compared without relying only on point estimates or null-hypothesis significance testing.

Each hypothesis receives its own measurement, its own model and its own estimand. Table 5 in Appendix D lists the models with the value each claim predicts, Table 1 every prior, and Table 3 the rule that turns a posterior into a verdict. All are ordinary regressions; only the quantity on the left and the question the coefficient answers change between them.

Each model is built the same way: a measurement, a linear predictor μ_i that adds one offset per group the observation belongs to, and a likelihood. One coefficient is the estimand and the rest keep it unconfounded. Table 1 gives every prior with the role its parameter plays. The range over which the prior scale is varied as a sensitivity check is $\{0.25, 0.5, 1.0\}$; all are centred on no effect, so a posterior that has not moved from its prior is itself a report of an absent effect.

H1

The behavioural measurement is the forecast amplitude recovery $R = A_{\text{pred}}/A_{\text{true}}$, collected in matched triplets: each lock f_k with its two controls, all three sharing the same background realisation and the same phase, which is what makes the contrast paired. The contrast used is

$$d = \log(R_k + 0.01) - \tfrac{1}{2}\left[\log(R_- + 0.01) + \log(R_+ + 0.01)\right], \tag{19}$$

one number per triplet, negative when the lock recovers less than its neighbours. It can be formed only because the injected tone has a known frequency. Model A fits it as

$$d_i \sim \text{Student-}t_4(\mu_i,\ \sigma), \quad \mu_i = \beta_{c[i]} + u_{k[i]} + u_{b[i]}, \quad \beta_c \sim \mathcal{N}\Big(\bar{\beta} + \delta_O \widetilde{O}_c + \delta_P \widetilde{\log P}_c,\ \tau\Big). \tag{20}$$

The estimand is $\bar{\beta}$, the population-level contrast at the average configuration. If locked frequencies are not attenuated relative to their neighbours, $\bar{\beta}$ is near zero and the posterior remains close to its prior; if they are, $\bar{\beta}$ is negative and its credible interval excludes zero.

- **Three offsets in μ_i**: β_c per configuration, u_k per lock harmonic and u_b per background draw, so that neither an unusual frequency nor a single corpus can be mistaken for the effect.
- **Student-t_4, not Gaussian.** Where the model rebuilds nothing, recovery is near zero and the logarithm returns very large contrasts; under a Gaussian a small number of them would determine the estimate.
- **β_c drawn around $\bar{\beta}$**, making the geometries instances of one phenomenon rather than unrelated experiments. Two consequences follow. $\bar{\beta}$ exists as a parameter, which $H1$ requires, being a claim about patch-based tokenisation and not about a single checkpoint; and each geometry contributes equal weight, so the largest design cannot dominate the estimate when the number of usable triplets varies several-fold with the in-band lock count.
- **Two configuration-level covariates**, estimable together rather than confounded because the design varies overlap and patch size independently. Overlap enters as a ratio, patch size as $\log P$ because the patch grid has spacing f_s/P and equal steps in $\log P$ are equal ratios of spacing. The tilde denotes centring on the design, $\widetilde{O}_c = (O_c - \bar{O})/0.5$ and $\widetilde{\log P}_c = \log P_c - \overline{\log P}$, which leaves $\bar{\beta}$ the effect at the average configuration rather than an extrapolation to $O = 0$ and $P = 1$.

Model B poses the same question of the representation rather than the forecast. At each frequency f_c a linear probe separates tones at $f_c \pm 1\,\mathrm{Hz}$ from a captured internal state, and the probe's prequential codelength ℓ measures how hard the separation is. Ten internal states are probed: the encoder [REG] token after each encoder block, the decoder state after each decoder block, the pre-projection vector and the quantile head. The regression is

$$\log \ell_i \sim \mathcal{N}(\mu_i,\ \sigma), \qquad \mu_i = \alpha + \theta_{\text{lock}} \cdot \text{IsLocked}_i + u_{s[i]} + u_{g[i]}, \tag{21}$$

where u_s and u_g are zero-mean offsets for probe stage and geometry, since codelengths differ several-fold between stages. The estimand is θ_{lock}: if locked frequencies are not harder to decode, θ_{lock} is near zero; if they are, $e^{\theta_{\text{lock}}}$ gives the factor by which the codelength increases. The band tasks summarise hundreds of frequencies at once, so the lock indicator has nothing to attach to; they are kept as a cross-check.

H2

$H2$ uses the same triplets with one change. Equation 19 has already averaged over phase, so the phase index is kept instead and the response becomes the per-phase deficit $y_i = \log(R_{\text{ctrl},i} + 0.01) - \log(R_{\text{lock},i} + 0.01)$, which is d with its sign reversed and positive when the lock is worse. Model C is Model A on that response, without the configuration-level covariates and with the phase circle cut into eight slices, each given its own offset u_p:

$$y_i \sim \text{Student-}t_4(\mu_i,\ \sigma), \qquad \mu_i = \beta_{c[i]} + u_{k[i]} + u_{b[i]} + u_{p[i]}, \qquad u_p \sim \mathcal{N}(0, \sigma_\phi). \tag{22}$$

The estimand σ_ϕ is the spread of those eight offsets. If the deficit does not depend on where in its cycle the signal starts, σ_ϕ is near zero, as $H2$ predicts; if it does, the offsets differ and σ_ϕ is large. Slices are used

Model	Parameter	Prior	Role
A, C	$\bar{\beta}$	Student-$t_4(0, 0.5)$	population phase-lock effect, on the log-recovery scale
A	δ_O	Student-$t_4(0, 0.5)$	slope in $\widetilde{O}_c$, the centred overlap; one unit $= 0.5$ of O
A	δ_P	Student-$t_4(0, 0.5)$	slope in $\widetilde{\log P}_c$, the centred log patch size
A, C	$\tau, \sigma_k, \sigma_b, \sigma$	Half-Student-$t_4(0, 0.5)$	configuration, harmonic, background and residual scales
C	σ_ϕ	Half-$\mathcal{N}(0, 0.25)$	spread of the eight per-phase offsets
B	α_0	$\mathcal{N}(\log \bar{y}, 1)$	baseline log codelength
B	θ_{lock}	$\mathcal{N}(0, 0.5)$	log codelength expansion at a lock
B	k	$\text{Gamma}(2, 0.1)$	Gamma shape (dispersion)
B	$\sigma_{st}, \sigma_{geo}$	Half-$\mathcal{N}(0, 1)$	spread of the probe-stage and geometry offsets
D1	$\alpha_g, \theta_S, \theta_P$	$\mathcal{N}(0, 1)$	per-geometry off-grid level and the two grid labels
D1	σ	Half-$\mathcal{N}(0, 1)$	residual scale of $\log z_g(f)$
D2	κ_S, κ_P	$\mathcal{N}(0, 1)$	measured spacing per unit of predicted spacing
D2	σ_F	Half-$\mathcal{N}(0, 5)$ Hz	scatter beyond the sweep-resolution floor Δf

Table 1: *Every parameter with its prior and its role. The effect priors are those of Deliverable 1, extended by δ_P, which the design of Table 2 makes estimable.*

rather than a fitted curve because they assume nothing about the form a dependence might take. Dropping u_p gives a nested null, and comparing the two by leave-one-out cross-validation (LOO) separates the size of the phase effect from the evidence that it exists.

H3

$H3$ is stated as two claims, one per branch. Both are tested on the token collapse $z_g(f)$, the dispersion of the input-patch embeddings across patches, which is zero when consecutive patches are identical. All geometries share one sweep grid: a uniform 1 Hz sweep unioned with the predicted sites of every configuration swept. No geometry is therefore scored only where its own theory predicts a dip, and non-integer sites such as $42.\overline{6}$ Hz for S=12 are covered.

The first question is where the dips are located. Each swept frequency carries two labels, one per branch:

$$\log z_g(f) \sim \mathcal{N}(\alpha_g + \theta_S \cdot \text{OnStrideGrid} + \theta_P \cdot \text{OnPatchGrid},\ \sigma)\,. \tag{23}$$

The two label coefficients θ_S and θ_P are the estimands. If the collapse dips do not align with the predicted grids, both coefficients are near zero; if they do, $H3$ predicts both negative, and the two-label fit should win the LOO comparison against each label alone and against neither. The two labels are not competing explanations of the same dips, since a frequency may carry either or both. The two coefficients separate, however, only where exactly one label applies, which for θ_S means the geometries with $S \nmid P$.

The second question is whether the dips move. The fit above remains compatible with dips that merely sit on a fixed grid, so each branch is checked against its own scaling law. Every detected dip is assigned to the branch that predicts it, and sites belonging to both are set aside. The fundamental $\hat{f}^F_{1,g}$ of branch F in geometry g is then compared with the spacing that branch predicts, $\Delta^S_g = f_s/S_g$ or $\Delta^P_g = f_s/P_g$:

$$\hat{f}^{F}_{1,g} \sim \mathcal{N}\left(\kappa_F\, \Delta^{F}_{g},\ \sqrt{\sigma_F^2 + \Delta f^2}\right). \tag{24}$$

The estimand is κ_F, measured spacing per unit of predicted spacing. If the branch tracks its own arithmetic, $\kappa_F = 1$; if the dips do not move with the geometry, κ_F departs from unity. The scatter σ_F about it is given a floor Δf, the sweep step, since a spacing read off a discrete grid is no more precise than one step; without the floor, sites landing exactly on the prediction drive σ_F to zero and the posterior degenerates.

***H3a*, the stride branch** The prediction is $\kappa_S = 1$: the stride sites track f_s/S one for one, so a site that remains fixed while S changes refutes it. Identification needs a series that holds P fixed and varies S, which Table 2 supplies at P=16, 24 and 32.

***H3b*, the patch branch** The prediction is $\kappa_P = 1$ in the same way. Identification needs a series that holds S fixed and varies P, which Table 2 supplies at S=8, 12 and 16.

Signal Generation and Testing

The overlap between consecutive patches is defined as

$$O = \frac{P - S}{P}, \tag{25}$$

where P is the patch length and S is the stride. When $O = 0$ the patches tile the signal without redundancy; when $O > 0$ consecutive patches share $P - S$ samples.

Run	**Patch (P)**	**Stride (S)**	O	**tokens**	$S \mid P$	$P \mid S$
1	8	4	0.50	511	yes	**no**
2	8	8	0.00	256	yes	yes
3	16	4	0.75	509	yes	**no**
4	16	8	0.50	255	yes	**no**
5	16	12	0.25	170	**no**	**no**
6	16	16	0.00	128	yes	yes
7	24	8	0.67	254	yes	**no**
8	24	12	0.50	169	yes	**no**
9	24	16	0.33	127	**no**	**no**
10	24	20	0.17	102	**no**	**no**
11	24	24	0.00	85	yes	yes
12	32	8	0.75	253	yes	**no**
13	32	12	0.62	169	**no**	**no**
14	32	16	0.50	127	yes	**no**
15	32	20	0.38	101	**no**	**no**
16	32	24	0.25	85	**no**	**no**
17	32	32	0.00	64	yes	yes

Table 2: *The extended configurations retained for the analysis. All are retrained from scratch under the same* 100*k-step budget and from the same seed (*42*). The token count is the number of patch tokens at the* 2048*-sample training context.* $S \mid P$ *means every stride lock is also a patch null, and* $P \mid S$ *the converse; both hold only at* $P = S$*, where the two branches are the same set of frequencies.*

The experimental evaluation uses three synthetic signal generators. The **single-harmonic sinusoidal generator** produces a pure tone $x(t) = A\sin(2\pi ft + \phi)$ sampled at a specified rate; its simplicity isolates frequency-dependent effects from spectral interactions. The **Light TSMixup generator** is a controlled probing variant of the TSMixup augmentation method[6], replacing real datasets with a pool of sinusoidal signals mixed via a symmetric Dirichlet distribution; its pseudocode is listed in Appendix B. The **KernelSynth generator**, also from the Chronos augmentation procedure[6], combines kernel functions with random weights to produce signals with richer frequency content; its pseudocode and kernel bank are listed in Appendix C.

Experimental Protocol

All signals are sampled at $f_s = 512\,\text{Hz}$. Pure sinusoids are swept from 2 to 250 Hz in one-hertz steps at several non-redundant starting phases, so that every tone is fully representable under the Nyquist–Shannon theorem and a loss observed at a predicted lock frequency cannot be attributed to undersampling. A single seed (42) controls all stochastic generation, ensuring full reproducibility.

The complete experimental grid is built before any data are collected: one row per combination of geometry (P, S), lock frequency, background type, and phase. For each experimental condition, 200 background draws from both the Light TSMixup and KernelSynth generators are archived alongside the probing results, so that the exact inputs behind every posterior can be reloaded for verification. Each geometry is probed independently: the retrained Chronos-Bolt weights are loaded, the synthetic signals are generated, predictions are run, and the outputs are written to the observation tables that feed the Bayesian models of Bayesian Analysis.

The geometries listed in Table 2 are chosen to allow independent variation of P and S. Series at fixed P and varying S (P=16 with $S \in \{4, 8, 12, 16\}$; P=24 with $S \in \{8, 12, 16, 20, 24\}$; P=32 with $S \in \{8, 12, 16, 20, 24, 32\}$) isolate the stride effect, while series at fixed S and varying P (S=8 with $P \in \{8, 16, 24, 32\}$; S=12 with $P \in \{16, 24, 32\}$; S=16 with $P \in \{16, 24, 32\}$) isolate the patch effect. This factorial structure is what lets Models A and D of Bayesian Analysis separate the two branches and estimate the overlap and patch-size covariates without confounding.

The design constants—sampling rate, analysis band, context window, prediction horizon and the retrained geometries of Table 2—are shared across all stages of the analysis. The Bayesian sampler uses the NUTS algorithm with 4 independent chains, each discarding 2000 warm-up draws and retaining the subsequent 2000 posterior samples; `target_accept` = 0.9 controls the step-size adaptation.

Decisions and Validation

Every threshold in Table 3 is fixed before a posterior is inspected, and validation comes before interpretation. Models A and C include a per-background random offset u_b with scale σ_b; this lets the posterior separate generator-specific variation from the aliasing effect itself, so that a recovery difference between Light TSMixup and KernelSynth backgrounds is attributed to the generator rather than inflating or deflating the estimate of $\bar{\beta}$.

Experimental Workflow

The analysis sweeps pure sinusoids from 2 to 250 Hz in one-hertz steps, at a sampling frequency of $f_s = 512\,\text{Hz}$. For each tone, the frequency produced by Chronos is compared with the known input frequency. Pure sinusoids keep the test easy to interpret: no background spectral content can explain a loss at a predicted lock frequency. The band stops below the Nyquist limit $f_s/2 = 256\,\text{Hz}$, so every tone swept is fully representable under the Nyquist–Shannon theorem, as Patch-Based Tokenization: a Formal Description sets out, and a loss observed here cannot be attributed to undersampling.

Before any Chronos output is loaded, the design skeleton is constructed: one planned row per geometry, lock frequency, background realisation, generator and phase. The priors of Table 1, the decision thresholds of

Claim	**Rule**	**Prior**	**Reads as**
$H1$ supported	$\Pr(\bar{\beta} < \log 0.8 \mid D) \geq 0.95$	0.34	at least 20% attenuation at a lock
$H1$ refuted	$\Pr(\lvert\bar{\beta}\rvert < \log 1.1 \mid D) \geq 0.95$	0.14	any effect is smaller than 10%
$H2$ supported	$\Pr(\sigma_\phi < \log 1.1 \mid D) \geq 0.95$	0.30	phase moves the deficit by under 10%
$H3$ location	two-label fit wins LOO, $\theta_S, \theta_P < 0$	—	dips sit on both predicted grids
$H3a$, $H3b$	$\Pr(\lvert\kappa_F - 1\rvert < 0.1 \mid D) \geq 0.95$	0.05	that branch tracks its own parameter
$M1$ supported	$\Pr(\delta_O < 0 \mid D) \geq 0.95$, overlap term wins LOO	0.50	overlap reduces the deficit

Table 3: *Decision rules.* ***Prior*** *is the probability each rule already scored under the priors of Table 1: a posterior counts as evidence only in as much as it has moved away from that value. A claim is supported at* 0.95*, refuted at* 0.05*, and otherwise inconclusive.*

Table 3, and the prior-sensitivity ladder $\{0.25, 0.5, 1.0\}$ are declared at this stage.

A collection stage then loads, for each geometry, the retrained Chronos-Bolt weights, generates the synthetic signals, runs the predictions, and writes the observation tables. The measurements are:

- **Behavioural (forecast recovery):** the sinusoid is given as context, the median forecast is appended and fed back to 512 generated samples. Its dominant frequency is read as the largest DFT magnitude after removing the mean and skipping the zero-frequency bin, then averaged over phases. This reading records what the model outputs and feeds Models A and C.
- **Representational (spectral recovery):** the internal state is captured at ten probe stages (encoder blocks, decoder blocks, pre-projection and quantile head). A linear probe (standardisation, principal component reduction, logistic regression) is scored by cross-validation on a frequency-local task: whether $f - 1\,\mathrm{Hz}$ can be separated from $f + 1\,\mathrm{Hz}$. This reading records what the model still represents and feeds Model B.
- **Token collapse:** the across-patch embedding dispersion $z_g(f)$ is measured at each swept frequency for each geometry. This feeds Models D1 and D2.

Collection is checkpointed per geometry, so an interrupted run resumes from the last incomplete geometry without altering the statistical design.

The five statistical models (A, B, C, D1, D2) are then defined and fitted with the same NUTS settings. Before fitting the real observations, a parameter-recovery test is run on Model A: synthetic contrasts are generated at four known effect sizes (0, $\log 0.8$, $\log 0.5$, $\log 0.1$), and the posterior is checked against the known value to verify that the experimental design has sufficient statistical power.

The fitted posteriors are either sampled or reloaded from checkpoints. For each hypothesis the analysis reports the 95% credible interval of the estimand, the posterior probability of the decision threshold, and the ROPE probability. Where competing formulations exist, leave-one-out cross-validation selects the one that better predicts unseen data. The comparisons are:

- Model A across configuration-level covariate sets, testing whether overlap or patch size mitigates the effect (M1);
- Model B with and without probe-stage and geometry corrections (H1 representational);
- Model C with and without per-phase offsets (H2);
- Model D1 across four grid labellings—stride only, patch only, both, neither—testing which branch generates the collapse sites (H3a, H3b).

The final stage verifies that each posterior can be trusted before interpreting it. Convergence diagnostics require $\hat{R} < 1.01$, adequate bulk and tail effective sample size, and zero divergences. Posterior predictive checks compare data generated from Model A against the observed contrasts. Prior sensitivity refits Model A across the scale ladder and with a Gaussian likelihood replacing the Student-t_4; the H1 conclusion is reportable only if the

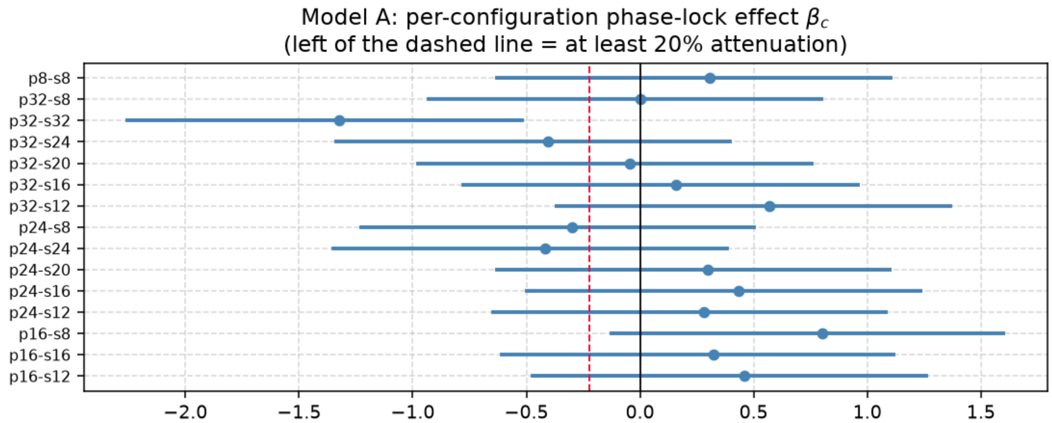

Figure 1: *Forest plot of the per-configuration phase-lock effect β_c (Model A, clean set). Each bar is the 95% credible interval; the dashed line at $\log 0.8$ marks 20% attenuation. The wide, bidirectional intervals explain the inconclusive population estimate.*

decision probability and credible interval remain stable. LOO comparisons apply the rule $|\text{elpd\_diff}| < 2 \cdot \text{dse}$ as inconclusive. A verdict table is written applying the three-way rule of Table 3: supported, refuted or inconclusive.

Behavioral and Representational Analysis

Bayesian Results

The models of Bayesian Analysis are fitted on two subsets of the grid (Table 2): a *clean* set of 15 configurations whose strides divide the probing context (CTX = 480), and a *full* set of 22 that adds seven geometries with shorter or odd strides. Posteriors are from the clean set unless stated otherwise; qualitative conclusions hold on both.

Validation

A parameter-recovery test on Model A fixes $\bar{\beta}$ at four known values (0, $\log 0.8$, $\log 0.5$, $\log 0.1$). Every 95% credible interval covers the truth, and the posterior standard deviation shrinks from the prior width of 0.71 to 0.04–0.18; the full set passes with tighter intervals (0.04–0.11).

Models D1 and D2 pass every convergence criterion ($\hat{R} \leq 1.003$, ESS > 1000, zero divergences). Models A and C do not ($\hat{R} > 1.3$, ESS < 12), and Model B falls short on ESS (676 bulk clean, 525 full). No posterior from a non-converged model is used to *support* a claim at the 0.95 threshold.

H1: Behavioural and representational effects

Behavioural (Model A) The population effect is $e^{\bar{\beta}} = 1.08$ [0.45, 2.49] (clean) and 1.02 [0.61, 1.70] (full). The probability of at least 20% attenuation is 0.19 and 0.21; the ROPE probability is 0.13 and 0.25: the verdict is **inconclusive**. Figure 1 shows the per-configuration posteriors spanning both sides of zero with no systematic direction.

Representational (Model B) The codelength expansion at a locked frequency is $e^{\theta_{lock}} = 1.27$ [1.23, 1.31] (clean, $\Pr > \log 1.2 = 1.00$, **supported**) and 1.22 [1.19, 1.24] (full, $\Pr = 0.84$, **inconclusive**). Both agree on direction and approximate size; the difference is whether the lower tail clears the 20% mark.

The split The behavioural test finds no consistent forecast attenuation; the representational test finds a consistent internal cost. This split — *representation penalised, forecast intact* — is the central finding: the model encodes a locked frequency less efficiently yet still reconstructs it, presumably because the decoder draws on enough redundant context to compensate.

H2: Phase invariance

The posterior is $\sigma_\phi = 0.033$ [0.020, 0.067] (clean) and 0.031 [0.019, 0.067] (full), well inside the ROPE at $\log 1.1 \approx 0.095$. The ROPE probability is 0.997 in both runs, against a prior of 0.30: the verdict is **supported**. Model C has convergence issues ($\hat{R} > 1.6$), but the margin is wide enough (0.997 vs 0.95) that the conclusion survives. Phase randomisation is therefore not a viable mitigation.

H3: Token-collapse sites

All D1 models converge ($\hat{R} \leq 1.003$, ESS > 5000). The stride-only label (M_S) wins LOO for TSMixup; the two-label model (M_{SP}) wins for pure sinusoids. In every comparison $\theta_S < 0$, confirming that frequencies on the stride grid have lower token dispersion. Figure 3 shows the dips sitting on the stride grid and shifting when the stride changes.

The stride scaling slope is $\kappa_S = 1.000$ [0.992, 1.008] (clean) and 1.006 [0.990, 1.022] (full), with $\Pr(|\kappa_S - 1| < 0.1) = 1.00$ in both runs against a prior of 0.048: $H3a$ is **supported**. The patch branch ($H3b$) is **not identified**: only 9 out of 180+ rows carry a valid fundamental, because in most configurations $S \mid P$ and no observation can be attributed to the patch branch alone.

Sensitivity and concordance

Model A is refitted across the prior-scale ladder {0.25, 0.5, 1.0} and with a Gaussian likelihood (Figure 4). The clean-set median of $\bar{\beta}$ moves from -0.05 to $+0.29$ to -0.31; the full set shows the same instability ($+0.06$ to -0.19 to $+0.17$). The 20%-attenuation probability ranges from 0.00 to 0.62 (clean) and 0.05 to 0.43 (full). The H1 behavioural conclusion is prior-dependent and cannot be reported as data-driven.

Where Bayesian and frequentist analyses address the same question, they agree: $H1$ behavioural inconclusive, $H2$ supported, $H3a$ supported. The well-converged models (D1, D2) deliver the firmest findings; the poorly converged ones (A, B, C) deliver qualified ones.

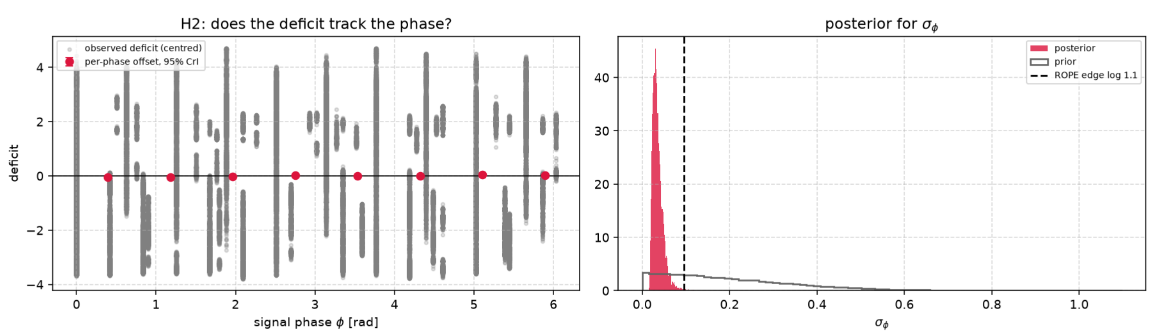

Figure 2: *Posterior for the phase term of $H2$. Left: observed deficit against phase ϕ with per-phase offsets u_p. Right: posterior for σ_ϕ against its prior, the dashed line marking the ROPE edge. Model C has not met convergence thresholds, so the posterior is provisional.*

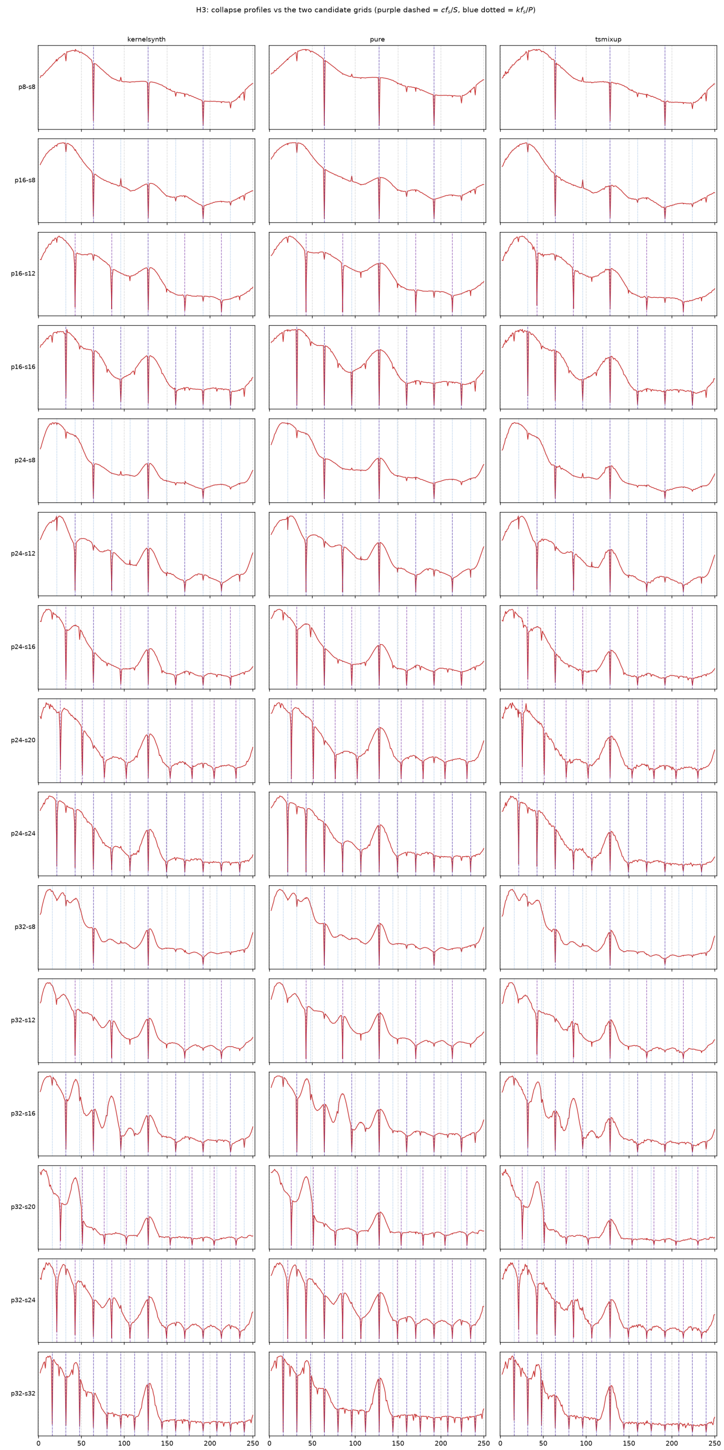

Figure 3: *Collapse profiles against the two candidate grids. Rows are configurations, columns generators; the stride grid cf_s/S is dashed and the patch grid kf_s/P dotted. Dips that follow the dashed grid as the stride changes belong to the stride branch.*

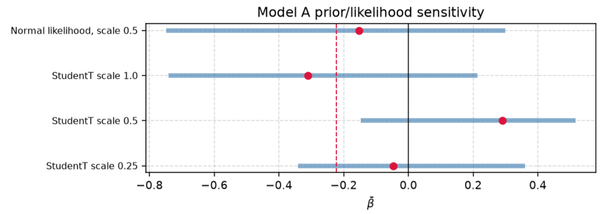

Figure 4: *Prior and likelihood sensitivity for Model A. The posterior of $\bar{\beta}$ is shown for three prior scales and for the Gaussian variant. The median shifts sign across variants.*

Mitigation Strategy

The mitigation pre-registered in Bayesian Analysis is increased patch overlap: shorter stride S at fixed P, raising $O = (P - S)/P$. The stride branch places its lowest in-band member at f_s/S, so halving S doubles that frequency and thins the family. The Bayesian test is Model A′ ($M1$ in Table 3), which fits the overlap-centred covariate δ_O.

The $M1$ test

The posterior of δ_O is 0.44 [−0.03, 0.81] (clean) and 0.38 [0.01, 0.72] (full). Both are *positive*: higher overlap is associated with a larger contrast, not a smaller one. The probability that $\delta_O < 0$ is 0.036 (clean) and 0.023 (full), far below the 0.95 threshold: the verdict is **refuted**.

Two interpretations are consistent. First, shortening S at fixed context multiplies the token count by P/S, changing the effective sequence length; this confound means the geometry effect and the sequence-length effect cannot be separated. Second, overlap thins only the stride branch while leaving the patch branch untouched, so the relative weight of the patch component grows and the aggregate contrast can move in the unexpected direction.

Arithmetic limits

Even setting the confound aside, the arithmetic bounds what overlap can achieve. The stride fundamental $f_{1,S} = f_s/S$ rises as S falls and can be thinned to any degree, but the patch fundamental $f_{1,P} = f_s/P$ depends on P alone: every integer multiple of f_s/P remains a potential lock site regardless of S. On the one-minute photovoltaic telemetry of the ontology domain ($f_s = 1/60\,\mathrm{Hz}$, $P = 16$), the patch harmonics have periods of 1 to 16 minutes and stay blind under any stride reduction.

Alternative directions

Three untested directions could address the patch branch: increasing P to raise $f_{1,P}$ and thin the patch family at the cost of coarser temporal resolution; a learnable tokeniser (e.g. a 1-D convolution with a trained stride) that steers its frequency nulls away from signal content; or a multi-resolution front end feeding tokens from two or more (P, S) pairs whose nulls fall at different frequencies.

The firmest practical conclusion is negative: an operator who increases the overlap ratio cannot rely on that change alone, because the component of the deficit generated by the patch grid is invariant to the stride.

Agentic AI

Computer Security

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Appendix A: Ontology of the Photovoltaic Application Domain

The core ontology architecture bridges the gap between cyber-physical energy systems and deep learning signal processing domain knowledge. It categorizes the application setting into five main conceptual pillars rooted under the ontological hierarchy:

- **Physical Subsystem (`pv:PhysicalComponent`):** Represents the hardware infrastructure of the microgrid, including the solar array, electrochemical storage, consumer loads, and the macro grid tie.
- **Cognitive Layer (`pv:ControlAgent`):** The artificial intelligence system responsible for autonomous grid optimization, load balancing, dispatch scheduling, and proactive vulnerability mitigation.
- **Signal Processing Layer (`pv:Signal`, `pv:PatchedSignal`):** Tracks the raw telemetry inputs (both time-domain features and frequency-domain components) along with their tokenized representation configurations (patch dimensions, stride values) tailored for the Chronos foundation architecture.
- **Environmental Tracking (`pv:WeatherObservation`):** Captures concurrent meteorological snapshots (air temperature, wind vectors) influencing structural physics and electrical efficiency.
- **Inferred Operational States:** Dynamic metadata classified into generation levels (`pv:GenerationState`), systemic disruptions (`pv:AnomalyState`), control directives (`pv:AgentAction`), and signal degradation hazards (`pv:AliasingEvent`).

Core Concept Taxonomy The formal classes defined within the schema are explicitly detailed below:

`pv:PhotovoltaicArray` The primary renewable energy asset. Characterized by panel count and nameplate nominal capacity. Source of telemetry streams; associated with weather conditions and subject to environmental/electrical anomalies.

`pv:BatteryStorage` Local electrochemical reservoir capable of dynamic grid connection/isolation. Characterized by capacity and instantaneous State of Charge (SoC).

`pv:InternalLoad` Represents internal industrial or residential consumption demands. Subject to adjustable, demand-side management interventions.

`pv:NationalGrid` The external electrical macro-grid interface, acting as a bidirectional infinite bus for excess export or deficit imports.

`pv:Signal` Time-series measurement stream containing global horizontal irradiance (G_h), tilted surface irradiance (G_{tilt}), instantaneous power output (P_{pv}), and analytical frequency descriptors (sampling rate, fundamental frequency, amplitude, phase).

`pv:PatchedSignal` The Chronos-specific tokenized structure derived from a raw signal. Governed by a selected window chunking mechanism (*patchSize*, *stride*), establishing a structural *overlapRatio* and a downstream $patchInducedFrequency$.

Semantic Relations & Properties

Object Properties (Entity-to-Entity Relationships) The physical, functional, and inferred linkages across concepts are formalized using the following object properties:

- ⟨`pv:PhotovoltaicArray`⟩ `pv:measures` ⟨`pv:Signal`⟩
- ⟨`pv:PhotovoltaicArray`⟩ `pv:hasWeatherObservation` ⟨`pv:WeatherObservation`⟩
- ⟨`pv:PhotovoltaicArray`⟩ `pv:hasGenerationState` ⟨`pv:GenerationState`⟩ *(Inferred via SWRL)*
- ⟨`pv:PhotovoltaicArray`⟩ `pv:hasAnomaly` ⟨`pv:AnomalyState`⟩ *(Inferred via SWRL)*
- ⟨`pv:PhotovoltaicArray`⟩ `pv:triggersAction` ⟨`pv:AgentAction`⟩ *(Inferred via SWRL)*
- ⟨`pv:Signal`⟩ `pv:tokenizes` ⟨`pv:PatchedSignal`⟩
- ⟨`pv:PatchedSignal`⟩ `pv:exhibitsAliasing` ⟨`pv:AliasingEvent`⟩ *(Inferred via SWRL)*
- ⟨`pv:BatteryStorage`⟩ `pv:buffers` ⟨`pv:PhotovoltaicArray`⟩
- ⟨`pv:PhysicalComponent`⟩ `pv:connectedTo` ⟨`pv:PhysicalComponent`⟩ *(Symmetric Property)*

Data Properties (Entity Attributes) Key quantitative attributes defining system conditions include:

- **Electrical/Solar:** `pv:pvPowerW` (P_{pv}), `pv:nominalPowerW` (P_{nom}), `pv:tiltedIrradianceWm2` (G_{tilt}), `pv:deltaPowerW` (ΔP).
- **Meteorological:** `pv:airTemperatureC` (T_{air}), `pv:windSpeedMs` (W_s).
- **Tokenization Architecture:** `pv:patchSize` (N_p), `pv:stride` (S), `pv:overlapRatio` (γ), `pv:patchInducedFrequency` (f_p).
- **Signal Descriptors:** `pv:samplingFrequencyHz` (f_s), `pv:fundamentalFrequencyHz` (f_0), `pv:phaseRad` (ϕ).

Knowledge-Driven Inference System (SWRL Rules) Real-time diagnostic, classification, and mitigation logic are automated using a rule-based engine. The standard logical form of these production rules is detailed below:

Environmental Context Extraction

$$\begin{aligned}&\texttt{PhotovoltaicArray}(?pv) \wedge \texttt{measures}(?pv, ?s) \wedge \texttt{tiltedIrradianceWm2}(?s, ?g) \wedge (?g \geq 800.0)\\ &\longrightarrow \texttt{hasGenerationState}(?pv, \texttt{ClearSky})\end{aligned} \tag{26}$$

$$\begin{aligned}&\texttt{PhotovoltaicArray}(?pv) \wedge \texttt{measures}(?pv, ?s) \wedge \texttt{tiltedIrradianceWm2}(?s, ?g)\\ &\wedge (?g \geq 150.0) \wedge (?g < 800.0) \longrightarrow \texttt{hasGenerationState}(?pv, \texttt{PartialCloud})\end{aligned} \tag{27}$$

$$\begin{aligned}&\texttt{PhotovoltaicArray}(?pv) \wedge \texttt{measures}(?pv, ?s) \wedge \texttt{tiltedIrradianceWm2}(?s, ?g)\\ &\wedge (?g > 0.0) \wedge (?g < 150.0) \longrightarrow \texttt{hasGenerationState}(?pv, \texttt{Overcast})\end{aligned} \tag{28}$$

$$\begin{aligned}&\texttt{PhotovoltaicArray}(?pv) \wedge \texttt{measures}(?pv, ?s) \wedge \texttt{tiltedIrradianceWm2}(?s, ?g) \wedge (?g \leq 0.0)\\ &\longrightarrow \texttt{hasGenerationState}(?pv, \texttt{Nighttime})\end{aligned} \tag{29}$$

Cyber-Physical Anomaly Diagnostics

- **Inverter Clipping Anomaly (R5):**

$$\begin{aligned}&\texttt{PhotovoltaicArray}(?pv) \wedge \texttt{measures}(?pv, ?s) \wedge \texttt{pvPowerW}(?s, ?p) \wedge \texttt{nominalPowerW}(?pv, ?nom)\\ &\wedge (?p \geq ?nom) \wedge \texttt{tiltedIrradianceWm2}(?s, ?g) \wedge (?g > 800.0)\\ &\longrightarrow \texttt{hasAnomaly}(?pv, \texttt{Clipping})\end{aligned} \tag{30}$$

- **Inverter Trip Anomaly (R6):**

$$\begin{aligned}&\texttt{PhotovoltaicArray}(?pv) \wedge \texttt{measures}(?pv, ?s) \wedge \texttt{pvPowerW}(?s, ?p) \wedge (?p \leq 0.0)\\ &\wedge \texttt{tiltedIrradianceWm2}(?s, ?g) \wedge (?g > 200.0) \longrightarrow \texttt{hasAnomaly}(?pv, \texttt{InverterTrip})\end{aligned} \tag{31}$$

- **Ramp Event Anomaly (R7):**

$$\texttt{PhotovoltaicArray}(?pv) \wedge \texttt{measures}(?pv, ?s) \wedge \texttt{deltaPowerW}(?s, ?dp) \wedge (?dp > 500.0) \longrightarrow \texttt{hasAnomaly}(?pv, \texttt{Ramp}) \tag{32}$$

- **Soiling Performance Deficit Anomaly (R8):**

$$\begin{aligned}&\texttt{PhotovoltaicArray}(?pv) \wedge \texttt{measures}(?pv, ?s) \wedge \texttt{tiltedIrradianceWm2}(?s, ?g) \wedge (?g > 400.0)\\ &\wedge \texttt{pvPowerW}(?s, ?p) \wedge (?p > 100.0) \wedge \texttt{swrlb:divide}(?r, ?p, ?g) \wedge (?r < 0.3)\\ &\longrightarrow \texttt{hasAnomaly}(?pv, \texttt{Soiling})\end{aligned} \tag{33}$$

- **Thermal Overheating Derating Anomaly (R9):**

$$\begin{aligned}\texttt{PhotovoltaicArray}(?pv) \wedge \texttt{hasWeatherObservation}(?pv, ?w) \wedge \texttt{airTemperatureC}(?w, ?t) \wedge (?t > 35.0)\\ \wedge \texttt{measures}(?pv, ?s) \wedge \texttt{tiltedIrradianceWm2}(?s, ?g) \wedge (?g > 600.0)\\ \wedge \texttt{pvPowerW}(?s, ?p) \wedge \texttt{swrlb:divide}(?r, ?p, ?g) \wedge (?r < 0.4)\\ \longrightarrow \texttt{hasAnomaly}(?pv, \texttt{Overheating})\end{aligned} \tag{34}$$

Autonomous Closed-Loop Control Actuation Based on inferred operating profiles, actions are dynamically mapped to handle grid stresses:

$$\texttt{pv:hasAnomaly}(?pv, \texttt{pv:Ramp}) \longrightarrow \texttt{pv:triggersAction}(?pv, \texttt{pv:RampAlert}) \tag{35}$$

$$\texttt{pv:hasAnomaly}(?pv, \texttt{pv:Clipping}) \vee \texttt{pv:hasAnomaly}(?pv, \texttt{pv:Overheating}) \tag{36}$$

$$\longrightarrow \texttt{pv:triggersAction}(?pv, \texttt{pv:CurtailGeneration}) \tag{37}$$

$$\texttt{pv:hasAnomaly}(?pv, \texttt{pv:InverterTrip}) \vee \texttt{pv:hasGenerationState}(?pv, \texttt{pv:PartialCloud}) \tag{38}$$

$$\longrightarrow \texttt{pv:triggersAction}(?pv, \texttt{pv:BatteryDispatch}) \tag{39}$$

$$\texttt{pv:hasGenerationState}(?pv, \texttt{pv:Overcast}) \vee \texttt{pv:hasGenerationState}(?pv, \texttt{pv:Nighttime}) \tag{40}$$

$$\longrightarrow \texttt{pv:triggersAction}(?pv, \texttt{pv:LoadAdjustment}) \tag{41}$$

Chronos Structural Aliasing Hypotheses The key scientific contribution of this architecture lies in identifying information loss bounds during time-series patching. The model detects structural blind spots via frequency domain reasoning:

Hypothesis H1 & H1b: Patch-Induced Frequency Blind Spots The patch tokenization process introduces an artificial operating frequency profile f_p, mathematically defined by the relationship:

$$f_p = \frac{f_s}{S} \tag{42}$$

Where f_s represents the raw Nyquist sampling rate and S denotes the token step stride.

The appropriate detection tolerance derives from the frequency resolution of the context window of length N_{context} samples:

$$\delta f \approx \frac{f_s}{N_{\text{context}}}. \tag{43}$$

For Chronos-Bolt default context lengths (512–2048 samples), this gives $\delta f \in [f_s/2048,\ f_s/512]$. The tolerance fraction $\varepsilon = \delta f/f_p = S/N_{\text{context}}$ must be validated empirically; whether $\pm 10\%$ is wider or narrower than this resolution depends on the specific S and N_{context} chosen.

- **Primary Blind Spot (H1):**

$$\begin{aligned}\texttt{Signal}(?s) \wedge \texttt{tokenizes}(?s, ?ps) \wedge \texttt{fundamentalFrequencyHz}(?s, ?f_0) \wedge \texttt{patchInducedFrequencyHz}(?ps, ?f_p)\\ \wedge \texttt{swrlb:divide}(?r, ?f_0, ?f_p) \wedge (?r \geq 1 - \varepsilon) \wedge (?r \leq 1 + \varepsilon)\\ \longrightarrow \texttt{exhibitsAliasing}(?ps, \texttt{H1\_BlindSpot})\end{aligned} \tag{44}$$

- **Secondary Harmonic Blind Spot (H1b):**

$$\begin{aligned}\texttt{Signal}(?s) \wedge \texttt{tokenizes}(?s, ?ps) \wedge \texttt{fundamentalFrequencyHz}(?s, ?f_0) \wedge \texttt{patchInducedFrequencyHz}(?ps, ?f_p)\\ \wedge \texttt{swrlb:divide}(?r, ?f_0, ?f_p) \wedge (?r \geq 2 - \varepsilon) \wedge (?r \leq 2 + \varepsilon)\\ \longrightarrow \texttt{exhibitsAliasing}(?ps, \texttt{H1b\_BlindSpot})\end{aligned} \tag{45}$$

Hypothesis H2: Phase Aliasing Effect Vulnerabilities appear when temporal offsets misalign with chunk window anchors, corrupting early representation extraction features ($0 < \phi < S$):

$$\begin{aligned} &\texttt{Signal}(?s) \wedge \texttt{tokenizes}(?s, ?ps) \wedge \texttt{phaseRad}(?s, ?\phi) \wedge \texttt{stride}(?ps, ?S) \\ &\quad \wedge (?\phi > 0.0) \wedge (?\phi < ?S) \longrightarrow \texttt{exhibitsAliasing}(?ps, \texttt{H2\_Phase}) \end{aligned} \tag{46}$$

Hypothesis H3: Overlap Information Distortion When the overlap configuration ratio ($\gamma = \frac{N_p - S}{N_p}$) drops below a critical information-theoretic bound ($\gamma < 0.5$), consecutive contextual patches display heavy text-token gaps, rendering trend forecasts highly unstable:

$$\texttt{PatchedSignal}(?ps) \wedge \texttt{overlapRatio}(?ps, ?or) \wedge (?or < 0.5) \longrightarrow \texttt{exhibitsAliasing}(?ps, \texttt{H3\_Overlap}) \tag{47}$$

Appendix B: Light TSMixup Pseudocode

Algorithm 1 Light TSMixup: Controlled Sinusoidal Time Series Mixup

Require: Sinusoidal pool $\mathcal{P} = \{p_1, \ldots, p_{N_p}\}$, where each p_n is defined by frequency f_n, amplitude A_n, phase ϕ_n, source length T_n, and sampling frequency f_s; maximum components $K = 10$; symmetric Dirichlet concentration $\alpha = 1.5$; minimum and maximum augmented lengths ($l_{\min} = 128, l_{\max} = 2048$).

Ensure: An augmented time series.

$k \sim U\{1, K\}$ ▷ number of sinusoidal components to mix

$l \sim U\{l_{\min}, l_{\max}\}$ ▷ length of the augmented time series

for $i \leftarrow 1$ to k **do**

  $n \sim U\{1, N_p\}$ ▷ sample a sinusoidal source

  $x_{1:l}^{(i)} \leftarrow \text{SampleSegment}(p_n, l, f_s)$ ▷ generate or crop a length-l segment

  $\tilde{x}_{1:l}^{(i)} \leftarrow x_{1:l}^{(i)} \Big/ \left(\frac{1}{l} \sum_{j=1}^{l} \left| x_j^{(i)} \right| \right)$ ▷ apply mean scaling

end for

$[\lambda_1, \ldots, \lambda_k] \sim \text{Dir}([\alpha_1 = \alpha, \ldots, \alpha_k = \alpha])$ ▷ sample mixing weights

return $\sum_{i=1}^{k} \lambda_i \tilde{x}_{1:l}^{(i)}$ ▷ take a weighted combination of scaled signals

Appendix C: KernelSynth Pseudocode and Kernel Bank

Algorithm 2 KernelSynth: Synthetic Data Generation using Gaussian Processes

Require: Kernel bank $\mathcal{K}$ listed in Table 4; maximum kernels per time series $J = 5$; generated length $l_{\text{syn}} = 1024$; optional numerical jitter $\epsilon = 0$.
Ensure: A synthetic time series $x_{1:l_{\text{syn}}}$.

```
j ∼ U{1, J}                                          ▷ sample the number of kernels
{κ_1(t, t′), ..., κ_j(t, t′)} ∼(i.i.d.) 𝒦               ▷ sample kernels with replacement
t ← linspace(0, 1, l_syn)
κ*(t, t′) ← κ_1(t, t′)                               ▷ initialize the composite covariance
for i ← 2 to j do
    ⋆ ∼ U{+, ×}                                      ▷ sample a binary composition operator
    κ*(t, t′) ← κ*(t, t′) ⋆ κ_i(t, t′)                ▷ compose kernels
end for
if ϵ > 0 then
    κ*(t, t′) ← κ*(t, t′) + ϵI
end if
x_{1:l_syn} ∼ 𝒢𝒫(0, κ*(t, t′))                       ▷ sample from the GP prior
return x_{1:l_syn}
```

Table 4: *Kernel bank $\mathcal{K}$ used by KernelSynth.*

Kernel	Formula	Hyperparameters
Constant	$\kappa_{\text{Const}}(x, x') = C$	$C = 1$
White Noise	$\kappa_{\text{White}}(x, x') = \sigma_n \cdot \mathbf{1}_{x=x'}$	$\sigma_n \in \{0.1, 1\}$
Linear	$\kappa_{\text{Lin}}(x, x') = \sigma_0^2 + x \cdot x'$	$\sigma_0 \in \{0, 1, 10\}$
RBF	$\kappa_{\text{RBF}}(x, x') = \exp\left(-\frac{\lVert x-x' \rVert^2}{2l^2}\right)$	$l \in \{0.1, 1, 10\}$
Rational Quadratic	$\kappa_{\text{RQ}}(x, x') = \left(1 + \frac{\lVert x-x' \rVert^2}{2\alpha}\right)^{-\alpha}$	$\alpha \in \{0.1, 1, 10\}$, $l = 1$
Periodic	$\kappa_{\text{Per}}(x, x') = \exp\left(-2\sin^2\left(\pi \frac{\lVert x-x' \rVert}{p/l_{\text{syn}}}\right)\right)$	$p \in \{24, 48, 96, 168, 336, 672,$ $7, 14, 30, 60, 365, 730,$ $4, 26, 52, 4, 6, 12, 4, 40, 10\}$

Model	Claim	Estimand	Predicts	Input
A	$H1$ behavioural	$e^{\bar{\beta}}$: recovery at a lock / at its controls	< 1	`contrasts`
B	$H1$ representational	$e^{\theta_{lock}}$: codelength at a lock / elsewhere	> 1	`mdl_cells`
C	$H2$	σ_ϕ: spread of the deficit across phase	≈ 0	`contrasts`
D1	$H3$ location	θ_S, θ_P: dip depth on each grid, by LOO	both < 0	`collapse`
D2	$H3a$ stride branch	κ_S: measured / stride-predicted spacing	$= 1$	`sites`
D2	$H3b$ patch branch	κ_P: measured / patch-predicted spacing	$= 1$	`sites`
A′	$M1$ mitigation	δ_O: change in the deficit per unit of overlap	< 0	`contrasts`

Table 5: *Summary of the Bayesian models, estimands, predicted values and probing inputs. This table is reproduced from the main text for convenience of reference alongside the full model specifications below.*

Appendix D: Bayesian Model Specifications

This appendix collects the full mathematical specification of each Bayesian model used in the analysis. The models are summarised in Table 5; the main text in Bayesian Analysis gives the high-level rationale and the estimands.

Model A (behavioural H1)

Model A tests whether the forecast amplitude recovery at locked frequencies is systematically lower than at neighbouring control frequencies. The measurement is the forecast amplitude recovery $R = A_{\text{pred}}/A_{\text{true}}$, collected in matched triplets: each lock frequency f_k paired with two controls $f_k \pm \delta$, sharing the same background realisation and the same phase. The log-contrast

$$d = \log(R_k + 0.01) - \tfrac{1}{2}\left[\log(R_- + 0.01) + \log(R_+ + 0.01)\right] \tag{19}$$

is negative when the lock recovers less than its neighbours. The offset 0.01 prevents the logarithm from diverging when recovery is near zero. Model A fits the contrast as

$$d_i \sim \text{Student-}t_4(\mu_i,\ \sigma), \quad \mu_i = \beta_{c[i]} + u_{k[i]} + u_{b[i]}, \quad \beta_c \sim \mathcal{N}\Big(\bar{\beta} + \delta_O \widetilde{O}_c + \delta_P \widetilde{\log P}_c,\ \tau\Big). \tag{20}$$

The linear predictor μ_i contains three offsets: β_c absorbs the per-configuration baseline, u_k absorbs variation across lock harmonics, and u_b absorbs variation across background draws. The Student-t_4 likelihood is chosen over a Gaussian because near-zero recoveries produce very large log-contrasts that would otherwise dominate the estimate. The configuration effects β_c are drawn around the population mean $\bar{\beta}$, making the geometries instances of one phenomenon; two centred covariates, overlap $\widetilde{O}_c$ and $\widetilde{\log P}_c$, allow the effect to vary with the configuration without confounding.

The estimand is $\bar{\beta}$. If locked frequencies are not attenuated, $\bar{\beta}$ is near zero and the posterior remains close to its prior. If they are attenuated, $\bar{\beta}$ is negative and its credible interval excludes zero.

Model B (representational H1)

Model B tests whether the internal representation retains less information at locked frequencies than at unlocked ones. At each frequency f_c a linear probe separates tones at $f_c \pm 1\,\text{Hz}$ from a captured internal state, and the probe's prequential codelength ℓ measures how hard the separation is. Ten internal states are probed: the encoder [REG] token after each encoder block, the decoder state after each decoder block, the pre-projection vector and the quantile head. The regression is

$$\log \ell_i \sim \mathcal{N}(\mu_i,\ \sigma)\,, \qquad \mu_i = \alpha + \theta_{\text{lock}} \cdot \text{IsLocked}_i + u_{s[i]} + u_{g[i]}, \tag{21}$$

where u_s and u_g are zero-mean offsets for probe stage and geometry, since codelengths differ several-fold between stages.

The estimand is θ_{lock}. If locked frequencies are not harder to decode, θ_{lock} is near zero. If they are, $e^{\theta_{\text{lock}}}$ gives the factor by which the codelength increases at a locked frequency relative to an unlocked one.

Model C (phase H2)

Model C tests whether the starting phase of the input signal modulates the deficit at locked frequencies. The response is the per-phase deficit $y_i = \log(R_{\text{ctrl},i} + 0.01) - \log(R_{\text{lock},i} + 0.01)$, positive when the lock is worse. The phase circle is cut into eight slices, each given its own offset u_p:

$$y_i \sim \text{Student-}t_4(\mu_i,\ \sigma), \qquad \mu_i = \beta_{c[i]} + u_{k[i]} + u_{b[i]} + u_{p[i]}, \qquad u_p \sim \mathcal{N}(0, \sigma_\phi). \tag{22}$$

The estimand is σ_ϕ, the spread of the eight phase offsets. If the deficit does not depend on phase, the offsets are equal and σ_ϕ is near zero, as H2 predicts. If it does depend on phase, the offsets differ and σ_ϕ is large. Slices are used rather than a fitted curve because they assume nothing about the functional form of a phase dependence.

Model D1 (collapse sites, H3 location)

Model D1 tests whether the token-collapse dips fall on the frequency grids predicted by the stride and patch branches. The token collapse $z_g(f)$, the dispersion of the input-patch embeddings across patches, is measured at each swept frequency. Each frequency carries two binary labels indicating whether it sits on the stride grid or the patch grid:

$$\log z_g(f) \sim \mathcal{N}(\alpha_g + \theta_S \cdot \text{OnStrideGrid} + \theta_P \cdot \text{OnPatchGrid},\ \sigma)\,. \tag{23}$$

The logarithm makes the labels multiplicative on the per-geometry off-grid level α_g. The two label coefficients θ_S and θ_P are the estimands. If the collapse dips do not align with the predicted grids, both coefficients are near zero. If they do, H3 predicts both negative, and the two-label fit should win the LOO comparison against each label alone and against neither. The two coefficients separate only where exactly one label applies, which for θ_S means the geometries with $S \nmid P$.

Model D2 (scaling law, H3 movement)

Model D2 tests whether the collapse dips move with the geometry as each branch predicts. Every detected dip is assigned to the branch that predicts it, and sites belonging to both are set aside. The fundamental $\hat{f}^F_{1,g}$ of branch F in geometry g is compared with the spacing that branch predicts, $\Delta^S_g = f_s/S_g$ or $\Delta^P_g = f_s/P_g$:

$$\hat{f}^F_{1,g} \sim \mathcal{N}\left(\kappa_F\, \Delta^F_g,\ \sqrt{\sigma_F^2 + \Delta f^2}\right). \tag{24}$$

The estimand is κ_F, measured spacing per unit of predicted spacing. If the branch tracks its own arithmetic, $\kappa_F = 1$; if the dips sit on a fixed grid that does not move with the geometry, κ_F departs from unity. The scatter σ_F is given a floor Δf (the sweep step) because a spacing read off a discrete grid is no more precise than one step; without the floor, sites landing exactly on the prediction drive σ_F to zero and the posterior degenerates.